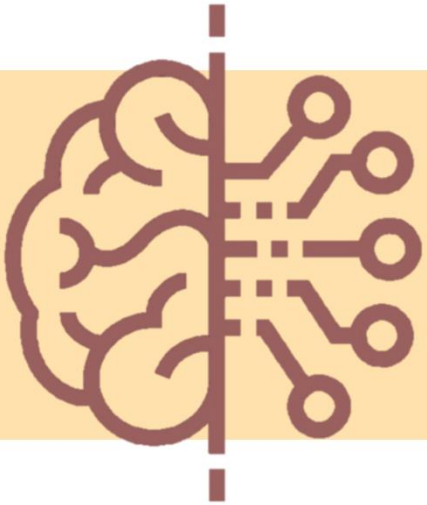
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Introduction to Computational Learning



Artificial Intelligence

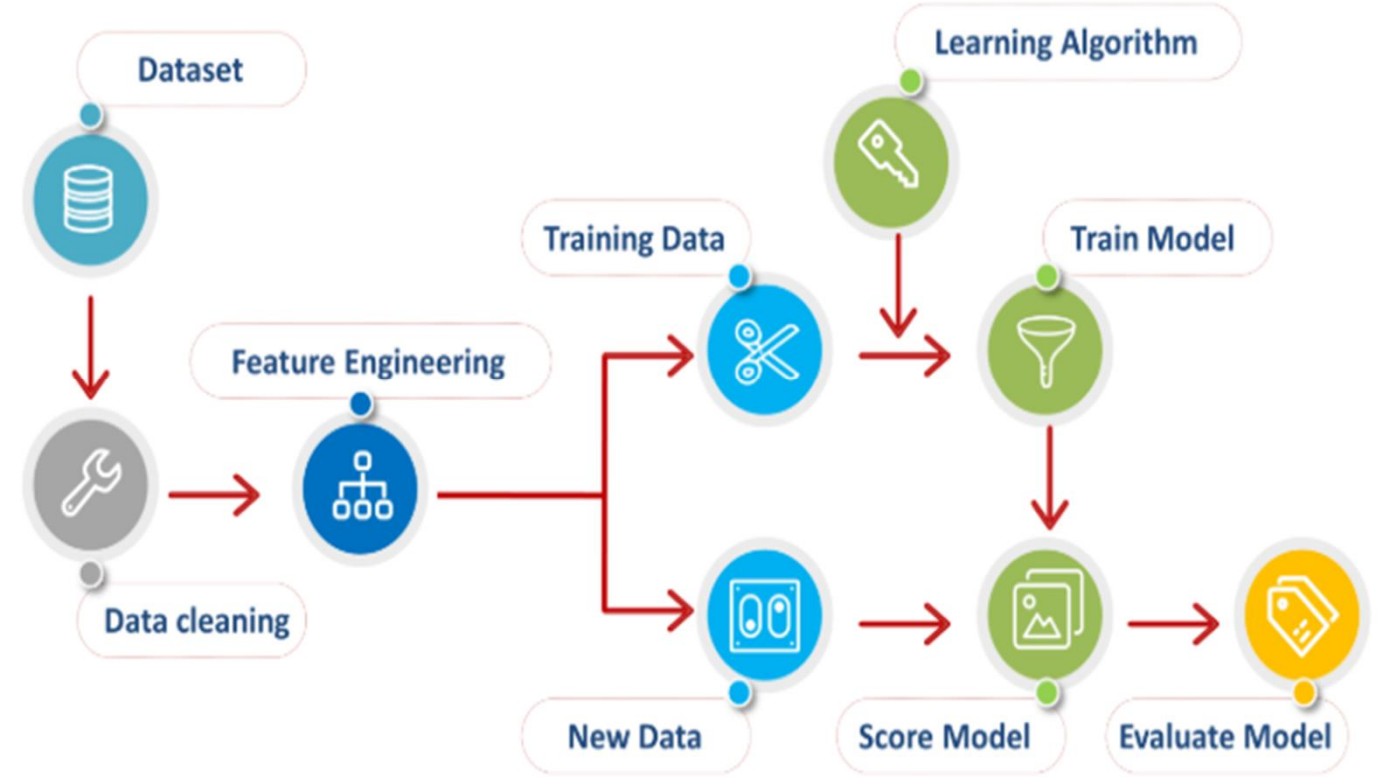
School of Computing
Universiti Teknologi Malaysia

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Machine Learning

How do we teach a machine to learn?

Machine Learning Process



How machine learns?

- Machine learning utilizes techniques of understanding **patterns** from input **data**
- Data is used for **training** by learning algorithm to **learn** and develop a specific computational **model**
- **Pre-processing** involves extracting important features of data
- **Model** refers to predefined prototype problem solver, produced based on patterns of input data

How good is the learning model?

- The aim of a model is to be able to produce good results
- Employ another set of data called **testing data** to **evaluate** the learning **model**
  - Can be completely new
  - Blinded dataset from the same resource
  - Smaller part of training data previously separated

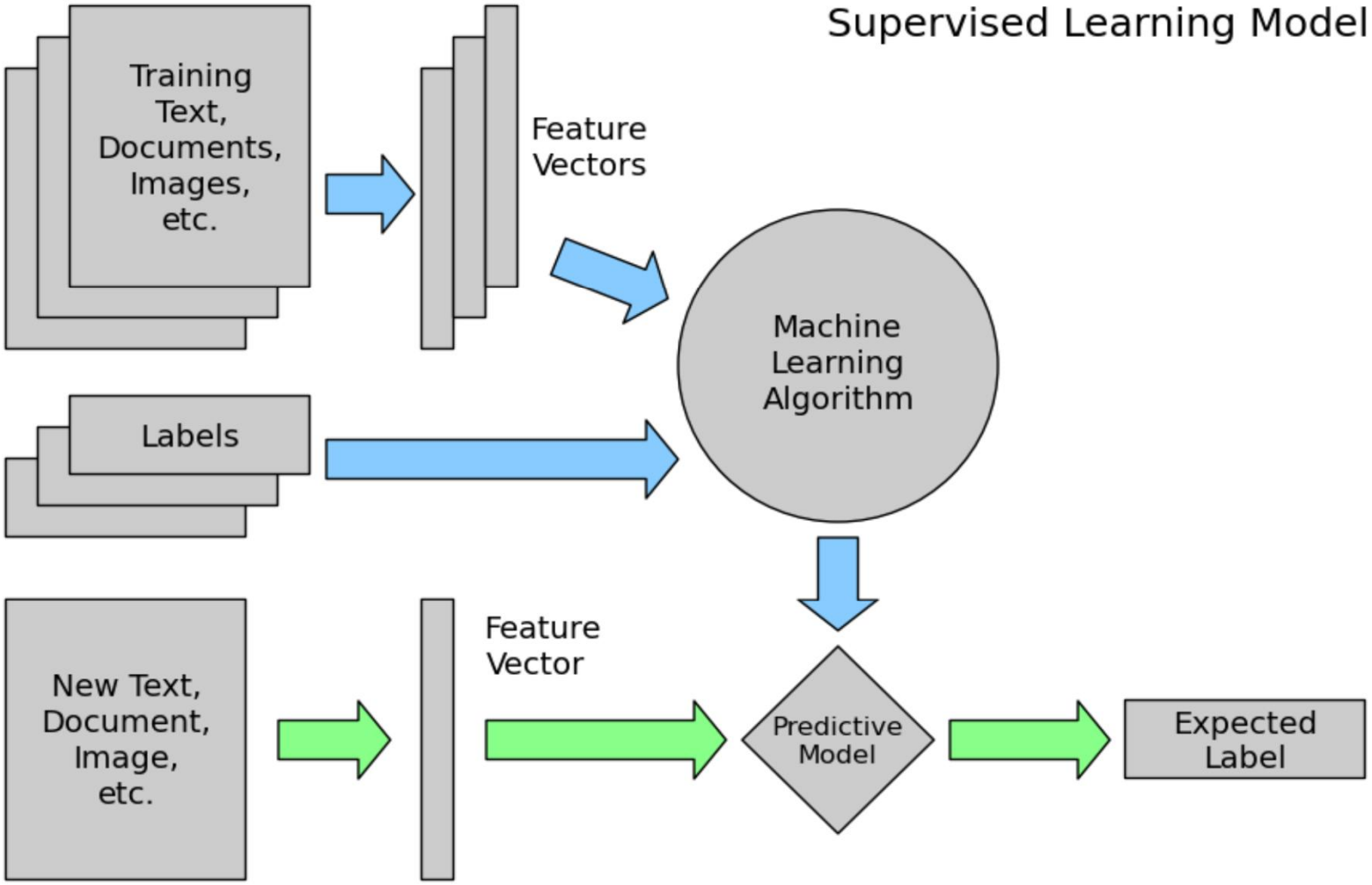
Learning types

- Supervised learning
  - The correct classes of the training data are known
- Unsupervised learning
  - The correct classes of the training data are NOT known
- Reinforcement learning
  - Machine learn its behavior based on feedback from environment

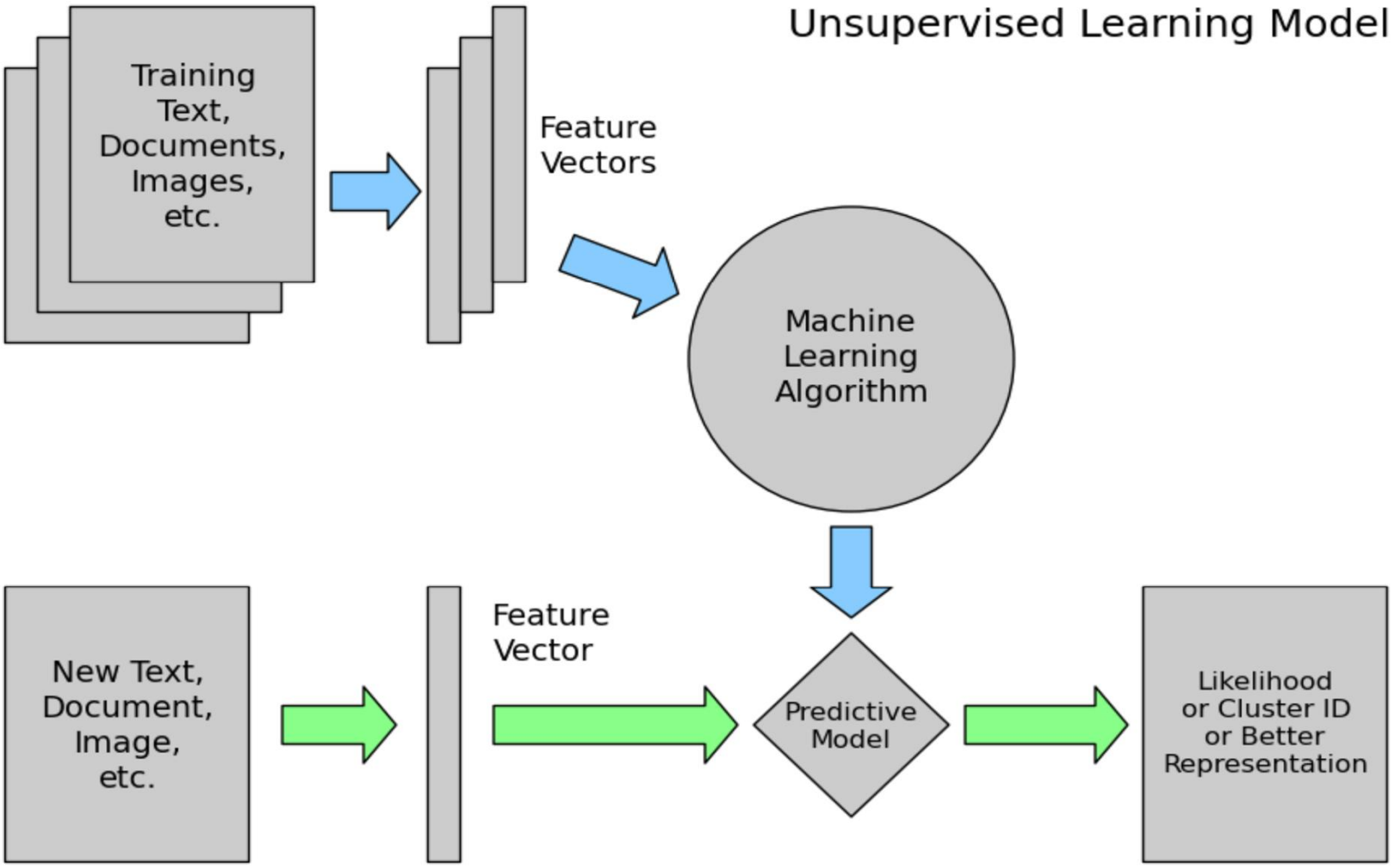
Learning types …

	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	The machine learns by using labelled data	The machine is trained on unlabelled data without any guidance	An agent interacts with its environment by producing actions & discovers errors or rewards
Type of problems	Regression & Classification	Association & Clustering	Reward based
Type of data	Labelled data	Unlabelled data	No pre-defined data
Training	External supervision	No supervision	No supervision
Approach	Map labelled input to known output	Understand patterns and discover output	Follow trail and error method
Popular algorithms	Linear regression, Logistic regression, Support Vector Machine, KNN, etc	K-means, C-means, etc	Q-Learning, SARSA, etc

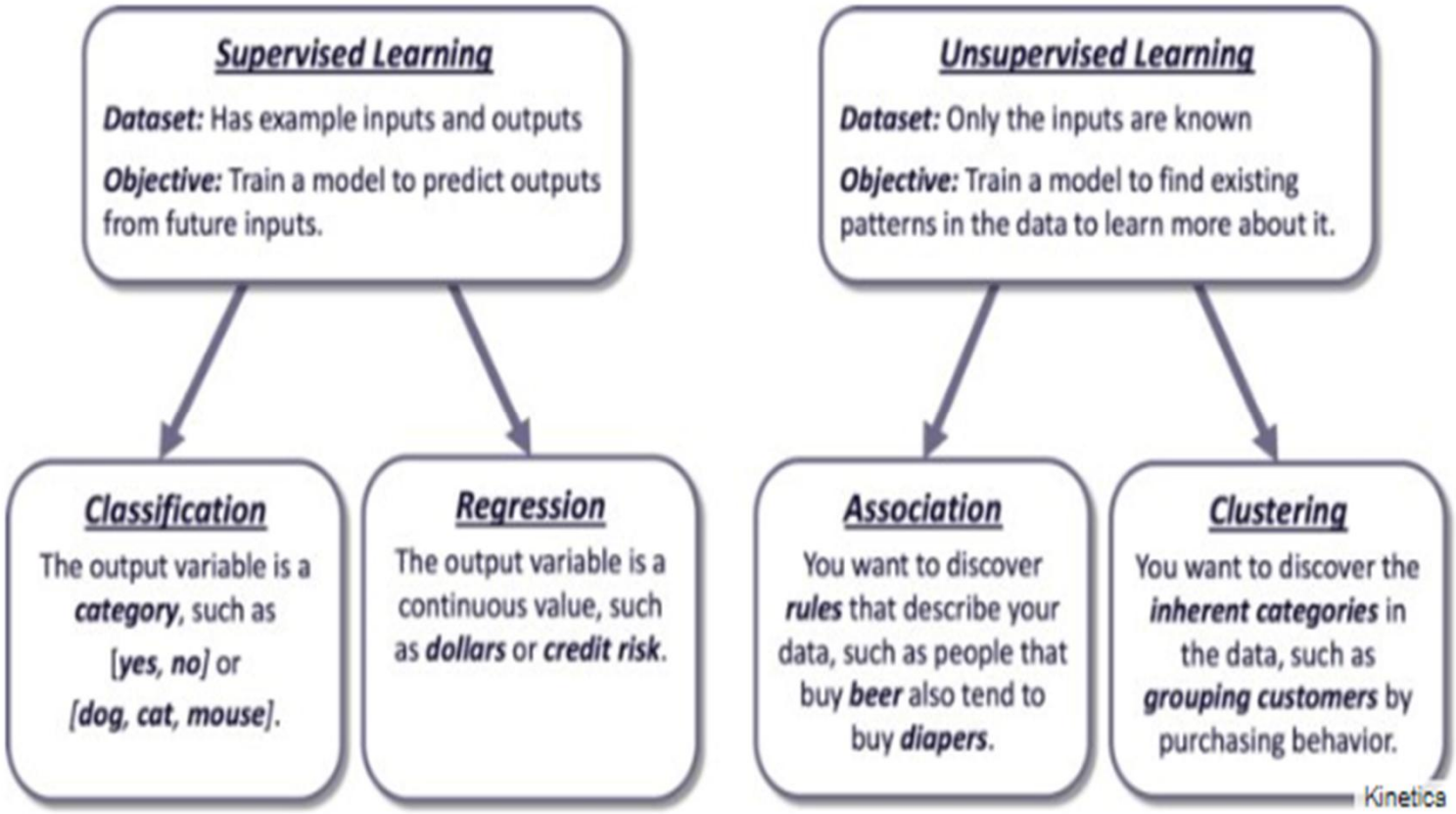
Supervised Learning



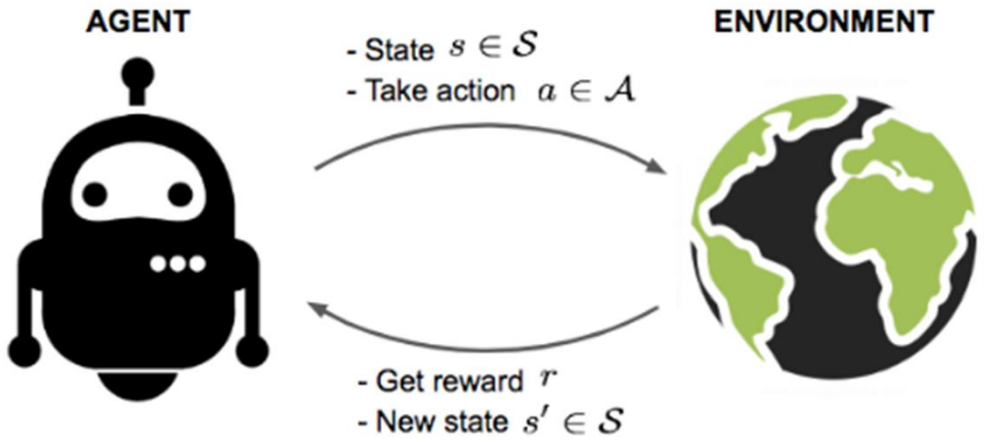
Unsupervised Learning



Supervised vs unsupervised

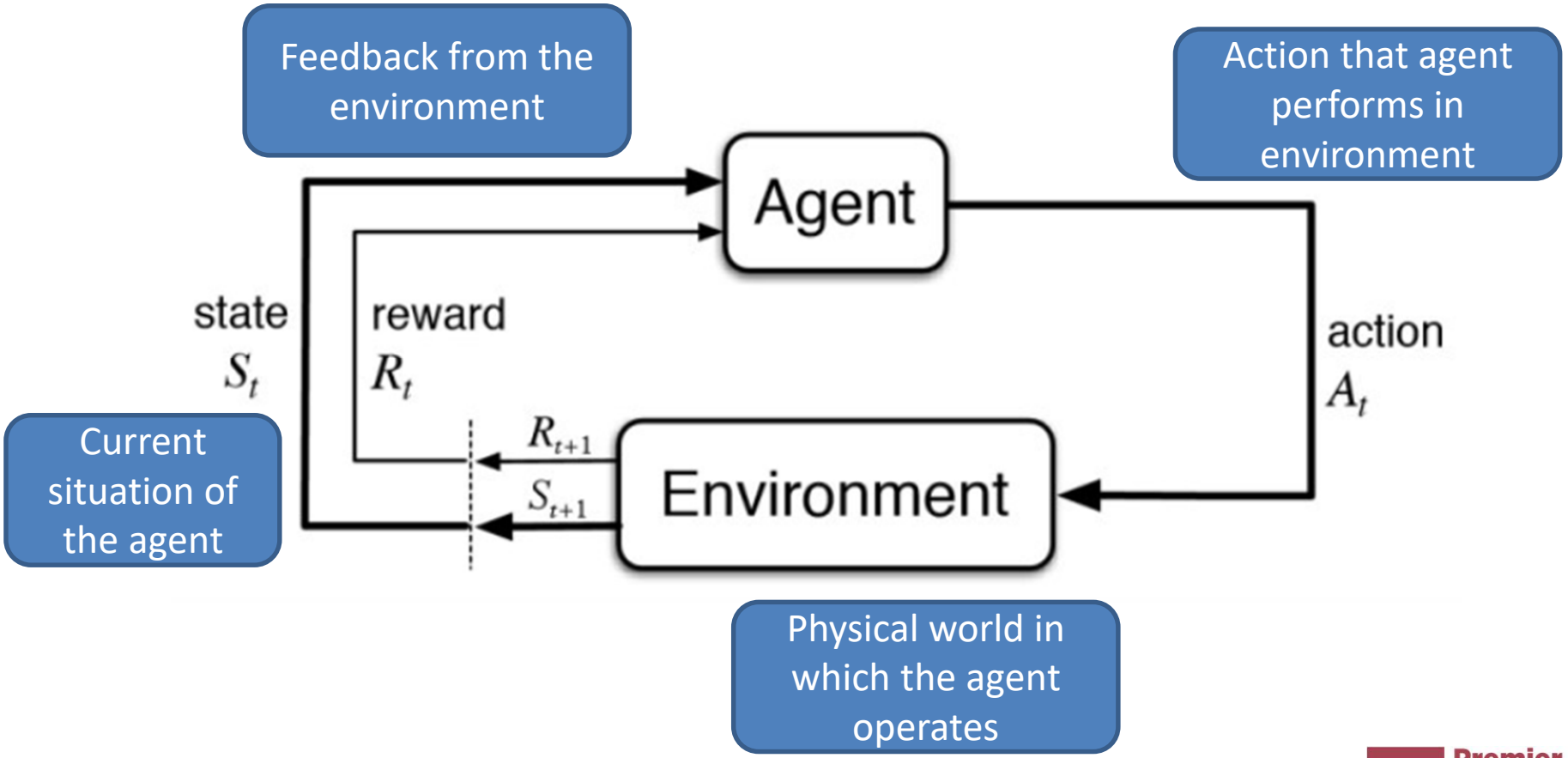


Reinforcement learning



- An agent interacts with the environment trying to take smart actions to maximize cumulative rewards
- The environment is formulated using Markov decision process to search for solutions

Markov Decision Process (MDP)



Markov Decision Process (MDP)

- The agent is acting in an **environment**
- How the environment reacts to certain actions is defined by a **model** which we may or may not know
- The agent can stay in one of the many **states** S of the environment and choose to take one of the many **actions** A so that it can switch from one state to another
- Which state the agent will arrive is determined by transition **probabilities** P between states
- Once an action is taken, a **reward** R is delivered by the environment as feedback

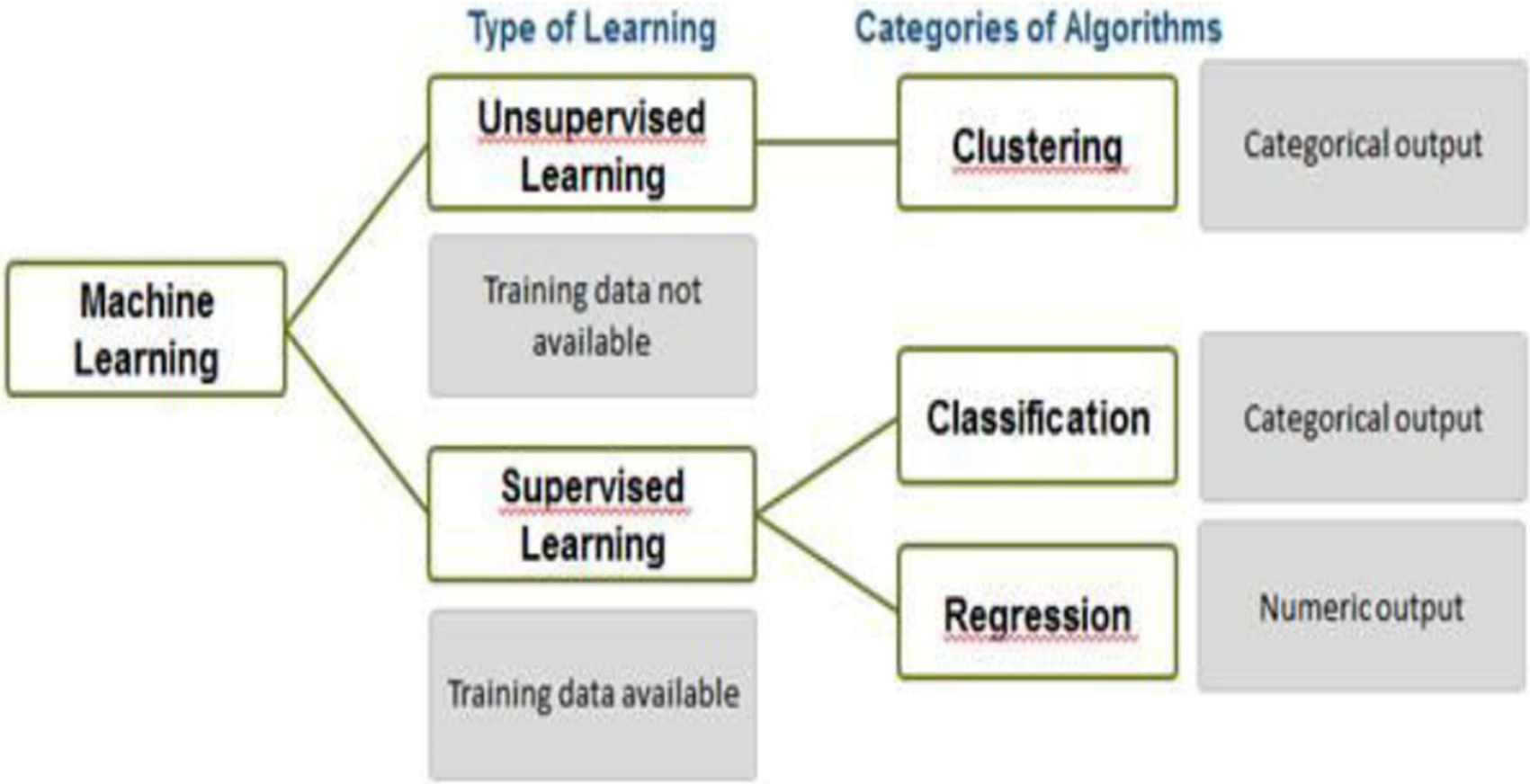
RL in real world

- Games
  - AlphaGo, AlphaGo Zero etc
- Robotic
  - (remember Spot from BostonDynamics?)
- Traffic light control
- Web system configuration
- And many more

Machine learning techniques

- Classification
  - Predict class from observation
- Clustering
  - Group observations into ‘’meaningful” groups
- Regression
  - Predict value from observations

Classification and clustering



How real world use ML?

https://towardsdatascience.com/real-world-examples-of-applied-machine-learning-from-ai-conference-4d4678700c6

Real world examples

- Uber COTA (Customer Obsession Ticket Assistant)

http://w4nderlu.st/assets/javascript/ViewerJS/#http://w4nderlu.st/content/3-publications/cota-improving-the-speed-and-accuracy-of-customer-support-through-ranking-and-deep-networks/cota-o-reilly-ai-conference-2018.pdf

- AirBNB ; categorizing listing photos

https://medium.com/airbnb-engineering/categorizing-listing-photos-at-airbnb-f9483f3ab7e3

- ZocDoc ; finding in network physicians from insurance cards

https://www.zocdoc.com/about/blog/tech/making-sense-of-insurance-cards-using-deep-learning/

- Spotify ; How does it know your kind of music?

https://medium.com/s/story/spotifys-discover-weekly-how-machine-learning-finds-your-new-music-19a41ab76efe

Computational Intelligence

What methods machines use to learn?

Outline

- What is AI?
- Now, come Computational Intelligence (CI)
- Scope of CI
- Importance of CI
- CI Techniques
  - Artificial Neural Network
  - Genetic Algorithm
  - Fuzzy Logic

What is AI?

- Ability to interact with the real world
- Reasoning and Planning
- Learning and Adaptation



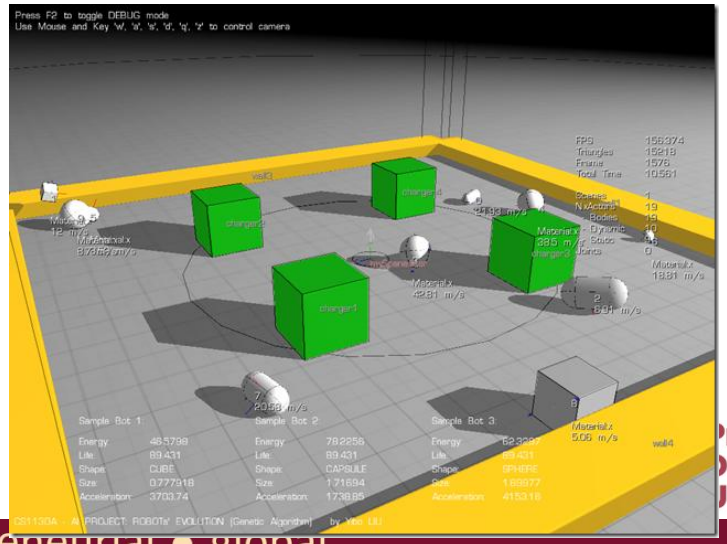
Now, come Computational Intelligence (CI)

- A methodology involving computing that exhibits an ability to learn and/or to deal with new situations, such that the system is perceived to possess one or more attributes of reason, such as generalization, discovery, association and abstraction (Russel et al, 1998).
- CI is the field of computing that draws from the successes of natural systems to develop new ways of solving computational problems in the real world.

Scope of CI

- practical adaptation concepts, paradigms, algorithms and implementations that enable or facilitate appropriate actions (intelligent behavior) in complex and changing environments.



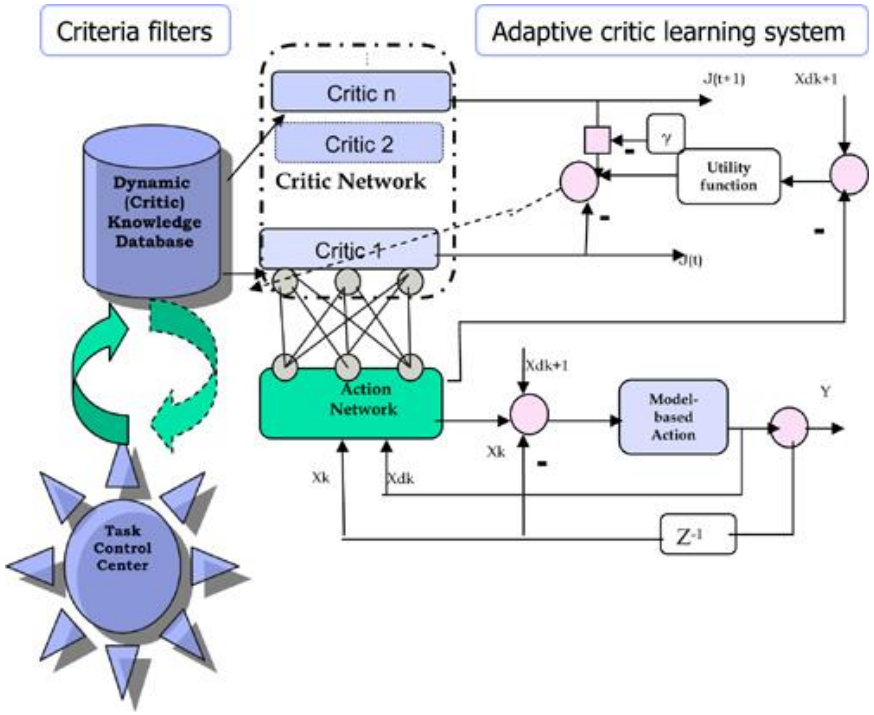


Scope of CI

- Fuzzy, imprecise or imperfect data
- No available mathematical algorithm
- Optimal solution unknown
- Rapid prototyping required
- Only domain experts available
- Robust system required
- Learning capability

More recently:

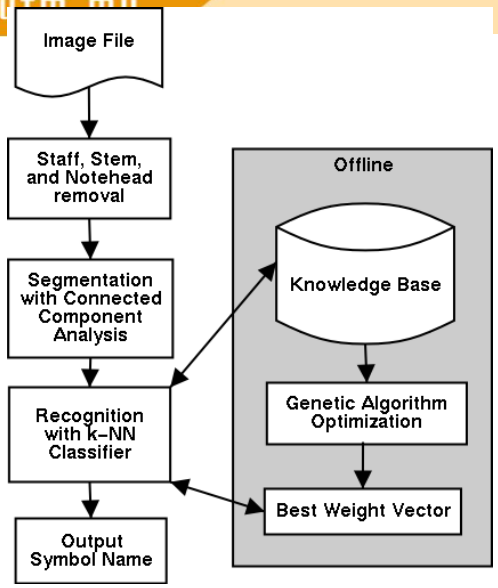
- DNA Computing
- Quantum Computing

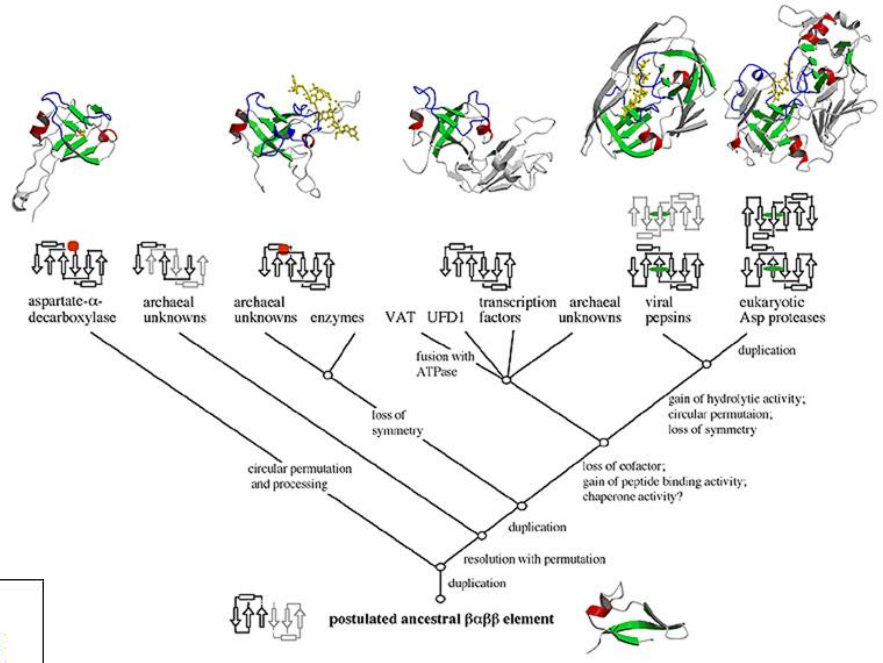


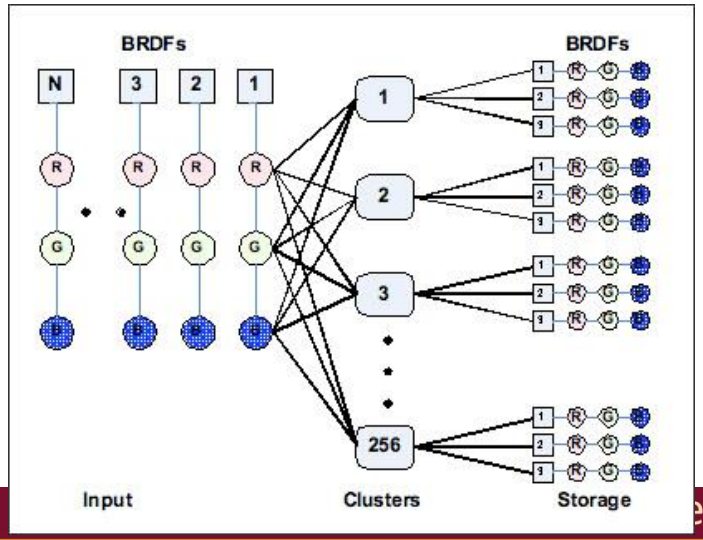
Importance of CI

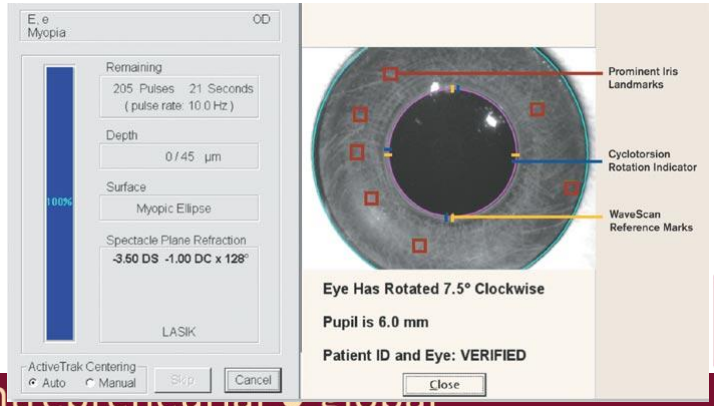
- Solve the problems in realistic situation (not ideal) such as fuzzy, imperfect or incomplete, only domain experts available.
- Complementary approaches with mathematical based algorithm solutions
- Encourage new technology development especially in Quantum and DNA computing.

Importance of CI





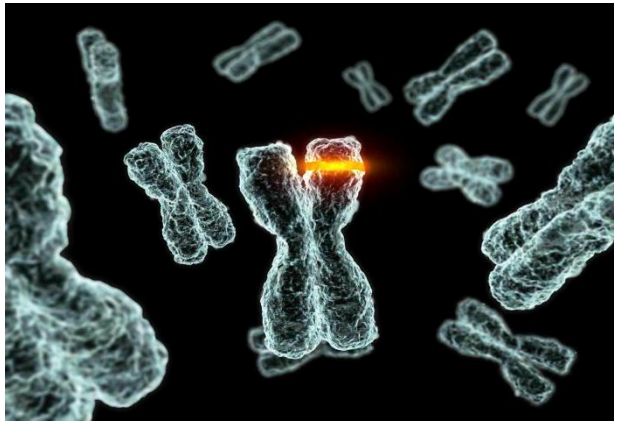
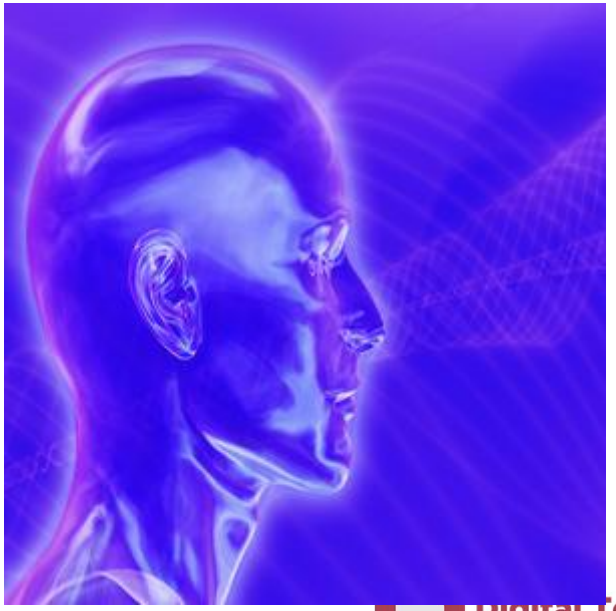




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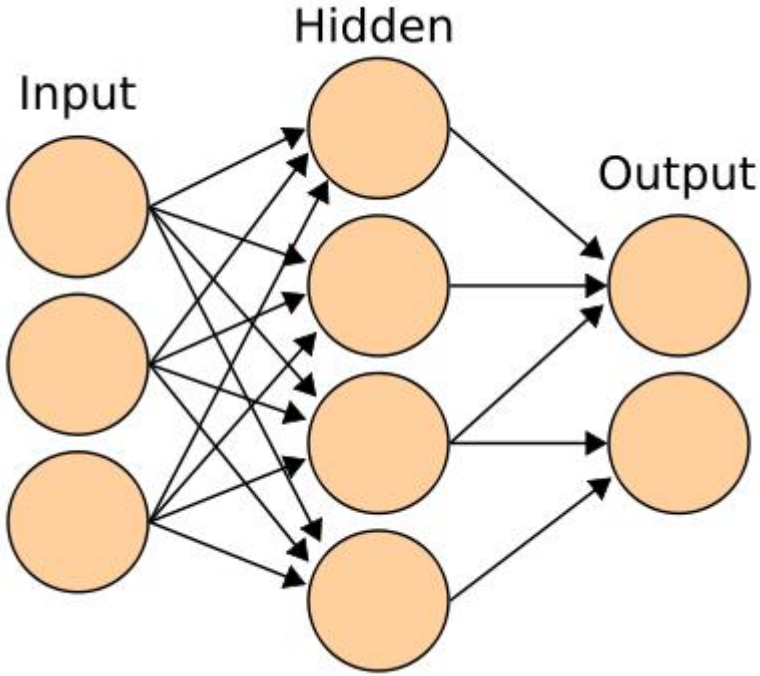
CI Techniques

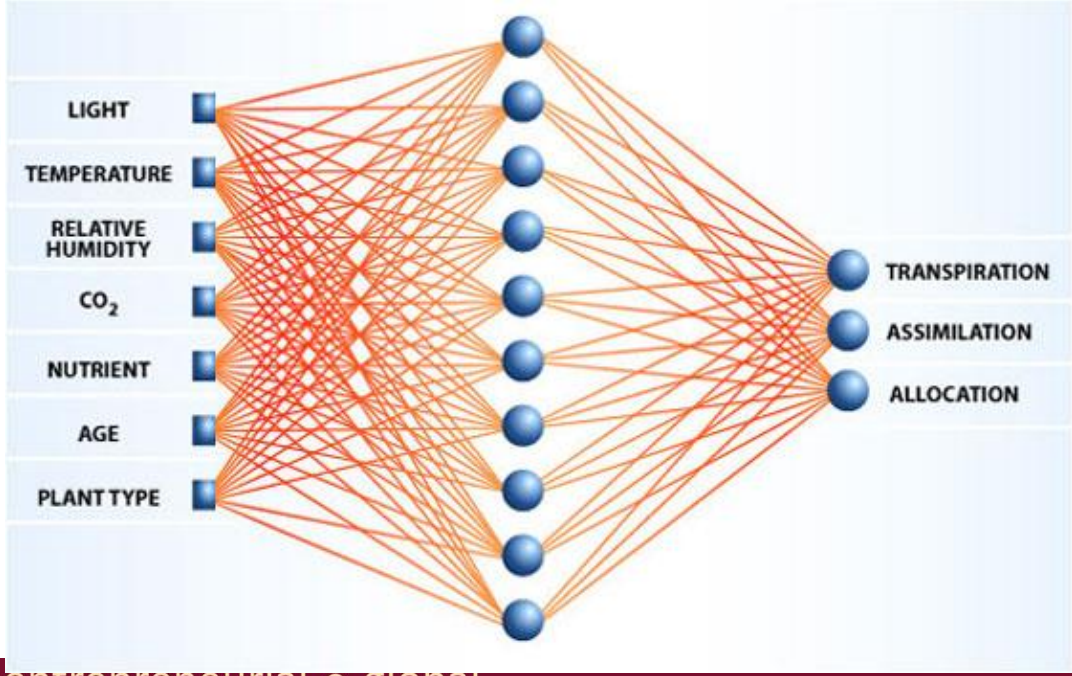
- Artificial Neural Network
- Genetic Algorithm
- Fuzzy Logic

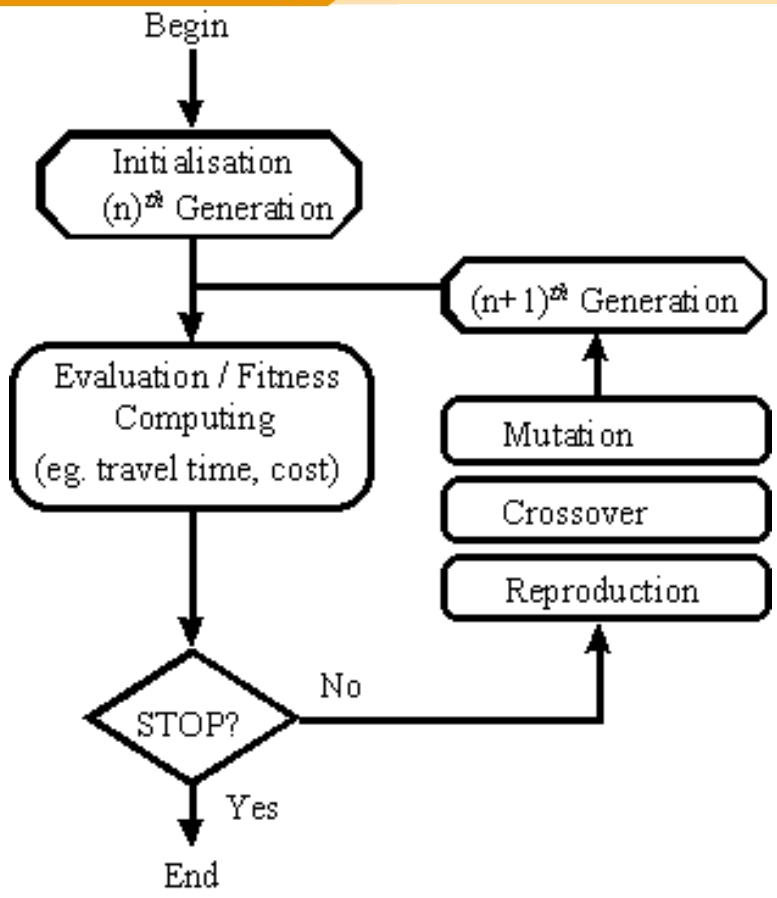
Artificial Neural Network







Genetic Algorithm



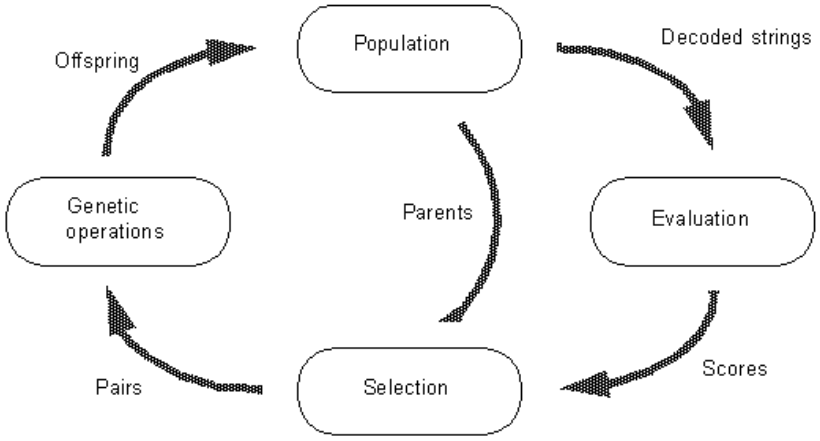
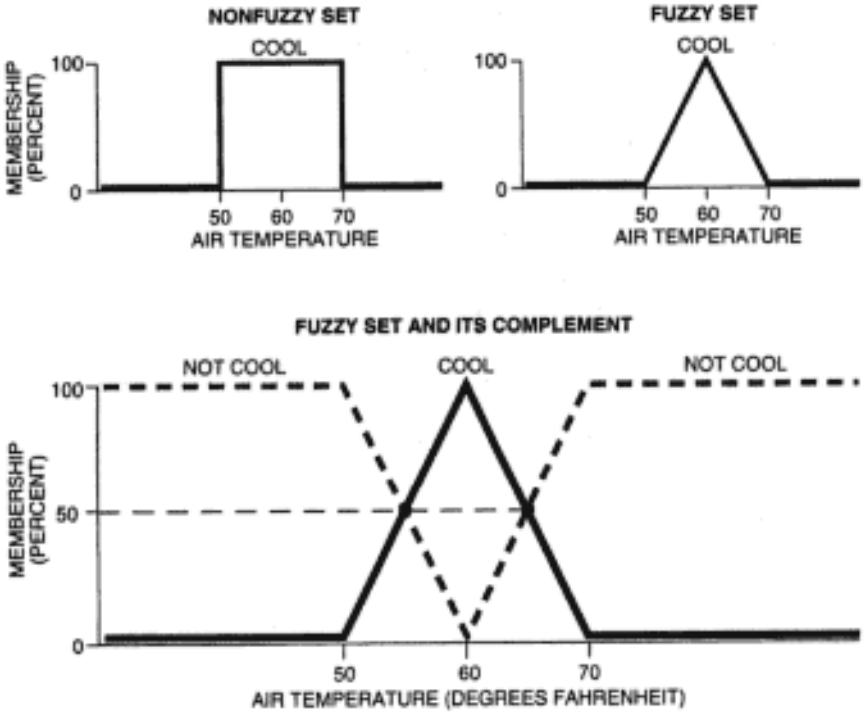
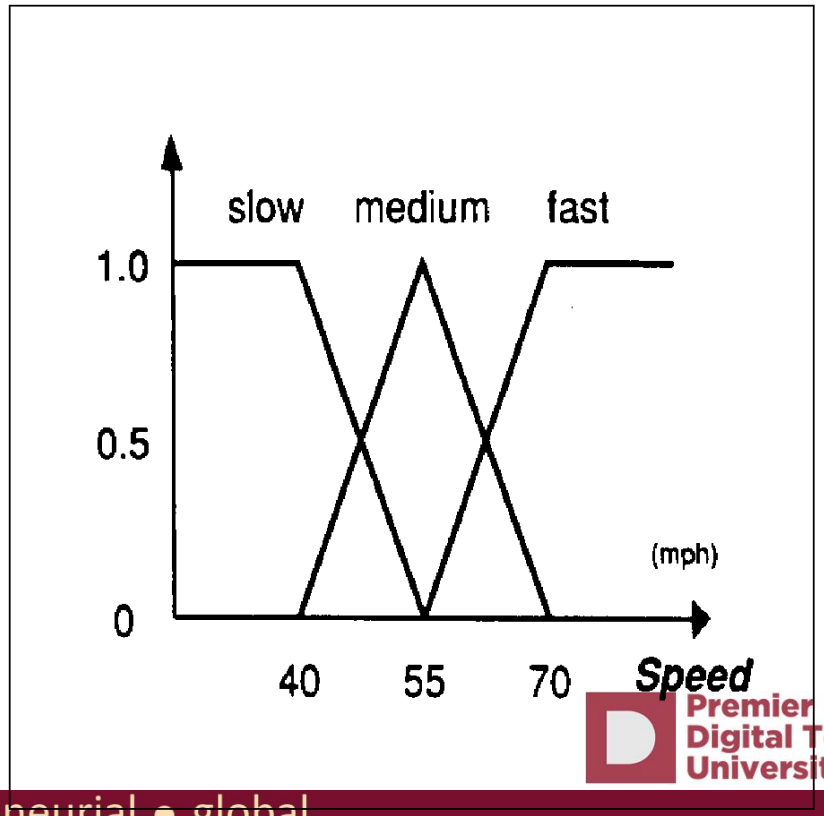


Figure 5.2: The "reproduction" cycle.

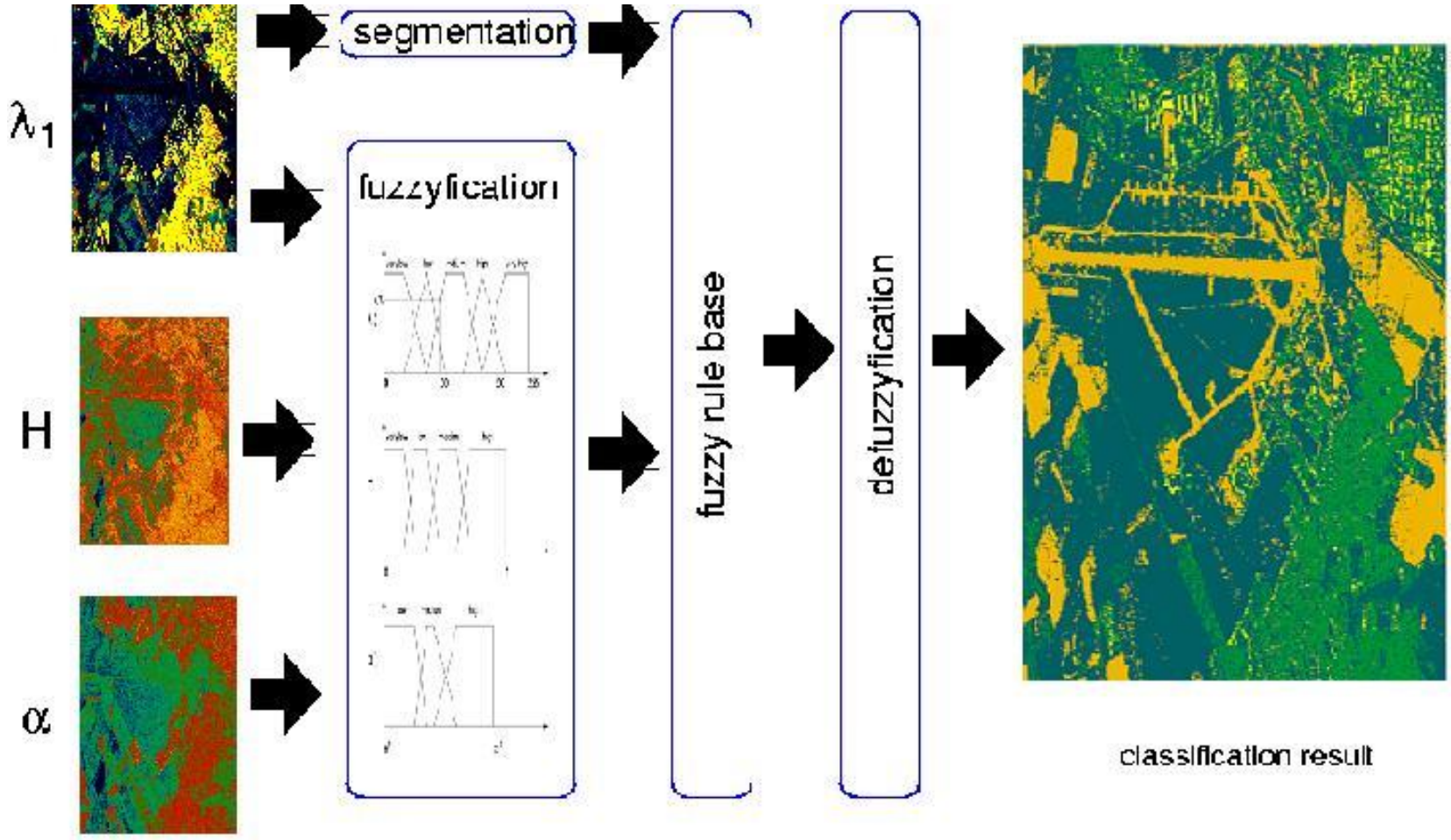


Fuzzy Logic


NONFUZZY SET
COOL
MEMBERSHIP (PERCENT)
100
0
50
60
70
AIR TEMPERATURE
FUZZY SET
COOL
100
0
50
60
70
AIR TEMPERATURE
FUZZY SET AND ITS COMPLEMENT
NOT COOL
COOL
NOT COOL
100
50
0
MEMBERSHIP (PERCENT)
50
60
70
AIR TEMPERATURE (DEGREES FAHRENHEIT)


slow
medium
fast
1.0
0.5
0
(mph)
40
55
70
Speed

Fuzzy Logic


λ1
H
α
segmentation
fuzzyfication
fuzzy rule base
defuzzyfication
classification result

Q & A

